

A Comparative Study of BVARs with Flexible Innovation Distributions

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1 Introduction

Overview

Bayesian Vector Autoregressions (BVARs) are widely used for modeling multiple time series in finance and macroeconomics. By placing priors on model parameters, BVARs provide a probabilistic framework for inference and forecasting that accounts for parameter uncertainty. In standard implementations, BVARs assume Gaussian innovations, which simplifies estimation and interpretation. However, this assumption is often violated in financial data, where returns exhibit heavy tails, volatility clustering, and structural changes over time. As a result, Gaussian-based models may fail to accurately capture the true data-generating process. This paper investigates extensions of the BVAR framework that incorporate flexible, non-Gaussian innovation distributions and evaluates their impact on predictive performance and risk estimation.

Problem Statement

The central challenge addressed in this work is the mismatch between Gaussian modeling assumptions and the empirical characteristics of financial returns. Standard Gaussian VAR models tend to underestimate extreme events and produce poorly calibrated predictive distributions, particularly during periods of market stress.

The key question is whether introducing more flexible innovation structures such as heavy-tailed distributions, regime-switching mechanisms, and time-varying volatility can improve both predictive accuracy and the modeling of tail risk. To address this, we compare four models within a unified framework: Gaussian VAR, Student- t VAR, mixture VAR, and stochastic volatility VAR.

Motivation

Empirical Features of Financial Returns

Financial return series exhibit several well-documented stylized facts. First, returns display heavy-tailed behavior, with extreme observations occurring more frequently than predicted by a Gaussian distribution. Second, volatility tends to cluster over time, with periods of high variability followed by similar periods. Third, returns across assets are strongly correlated, particularly during market downturns. These features are clearly observed in the ETF return data used in this study, which spans multiple market regimes including the 2008 financial crisis and the COVID-19 shock. Such empirical characteristics highlight the need for models that can capture both tail behavior and time-varying risk.

Limitations of Gaussian Assumptions

Gaussian VAR models impose thin-tailed innovations and constant covariance structures, which are often inadequate for financial applications. In particular, these models tend to underestimate the probability of extreme losses, leading to poor performance in risk-sensitive tasks such as Value-at-Risk and Expected Shortfall estimation. Additionally, Gaussian models cannot account for structural changes or regime shifts, which are common in financial markets. As a result, their predictive performance deteriorates during periods of heightened volatility or market stress.

Why Non-Gaussian BVARs

To address these limitations, we consider extensions of the BVAR framework that relax the Gaussian assumption. Heavy-tailed models, such as the Student- t VAR, provide robustness to outliers by allowing for more flexible tail behavior. Mixture models introduce regime-switching dynamics, capturing distinct market states such as normal and crisis periods. Stochastic volatility models allow the variance to evolve over time, capturing volatility clustering. These approaches target different aspects of model misspecification, and their comparative performance provides insight into which features are most important for accurate forecasting and risk modeling in financial time series.

2 Literature Review

We begin with the foundations of VAR modeling and then discuss BVAR frameworks and the role of shrinkage priors in improving estimation and forecasting performance. We next review extensions that relax the conventional Gaussian innovation assumption, with particular emphasis on Student- t innovations, mixture-based specifications, and stochastic volatility models. Finally, we summarize the main approaches to probabilistic forecast evaluation and metrics.

Foundations of Vector Autoregression

Vector autoregressions (VARs) constitute a fundamental framework for multivariate time-series analysis in macroeconomics and finance. To begin with, Sims [1] criticized the strong a priori restrictions imposed by traditional structural macroeconomic models and proposed a more data-driven approach. The main variables in this work are modeled through their lagged interdependence. This formulation established the VAR as a flexible empirical framework for characterizing dynamic interactions among economic variables. Its influence remains substantial in both applied forecasting and empirical macroeconomic analysis [2, 3].

However, a key challenge is that unrestricted VARs become rapidly overparameterized as the number of variables and lag length increase. Early work recognized that the practical success of VARs depends not only on their flexibility but also on the ability to control estimation uncertainty in high-dimensional data. In this regard, Doan et al. [4] made a major contribution by introducing Bayesian estimation and forecasting procedures based on realistic prior distributions. Their framework showed that prior regularization can preserve the multivariate dependence structure of VARs while improving forecasting performance. Litterman [5] further showed that Bayesian vector autoregressions can improve forecasting performance and helped popularize what later became known as the Minnesota prior. By shrinking coefficients toward simple dynamic structures, this approach controls overfitting while keeping the cross-variable linkages that are central to VAR models.

Recent work also shows that the VAR framework remains highly relevant once the focus moves beyond standard mean dynamics. For example, Chavleishvili and Manganelli [6] develop a quantile VAR framework for forecasting and stress testing, showing that VAR systems can be adapted to study the evolution of conditional quantiles and macroeconomic tail risk. In a different direction, Karlsson et al. [7] extend the VAR model to allow for skewness and heavy tails in the innovation distribution. Taken together, these contributions illustrate that VARs continue to serve as the baseline multivariate architecture for modern forecasting models, including heavy tails and tail-risk dynamics [8, 9].

Bayesian VAR and Shrinkage Priors

Although the VAR framework is flexible and widely applicable, its empirical performance often deteriorates when the system contains many variables or lags. In such settings, the number of coefficients grows quickly, making unrestricted estimation unstable and prone to overfitting. Bayesian vector autoregressions (BVARs) [4] were developed to address this issue by introducing prior distributions on the model parameters. Rather than estimating all coefficients freely, the Bayesian approach regularizes the model by shrinking parameters toward statistically plausible values, thereby improving estimation stability and forecasting performance [10, 11].

A central contribution in this work is the Minnesota prior, introduced in early work by Litterman [5]. The basic idea is to shrink coefficients on longer lags and cross-variable effects toward zero, while allowing a variable’s own recent lags to retain greater influence. This prior reflects the belief that many economic time series are persistent, but that distant lags and weak cross-series effects should not be estimated too freely in finite samples. In practice, the Minnesota prior became one of the most influential tools in Bayesian macroeconomic forecasting because it offers an effective balance between flexibility and regularization [12, 13].

Subsequent research extended these ideas to larger systems and a more systematic prior design [14, 15]. For example, Bańbura et al. [16] showed that, with appropriately calibrated shrinkage, large Bayesian VARs can incorporate broad information sets and still achieve strong forecasting performance. This result was especially important because it demonstrated that Bayesian regularization makes high-dimensional VAR forecasting feasible even when the number of parameters is very large. Later, Giannone et al. [17] studied the problem of prior selection more formally and proposed principled ways to choose hyperparameters in BVARs. Their work showed that successful Bayesian forecasting depends not only on using shrinkage but also on choosing its strength appropriately for the data.

More recent work continues to highlight the importance of shrinkage design in Bayesian VAR forecasting [18, 19]. For instance, Gruber and Kastner [20] shows that the predictive performance of a BVAR can vary substantially depending on whether the underlying dependence structure is sparse or dense. This finding suggests that prior specification should be chosen to match the structure of the data. Taken together, these studies establish BVARs as a natural extension of the classical VAR framework: by combining multivariate autoregressive structure with prior regularization, BVARs provide a practical and powerful approach to forecasting in high-dimensional financial environments.

Limitations of Gaussian Innovations

A common assumption in standard VAR models is that the innovation term follows a multivariate Gaussian distribution [21, 22]. This assumption is attractive because it simplifies both estimation and forecasting. However, it can be restrictive in many macroeconomic and financial applications [23, 24].

Empirical financial time series often display non-Gaussian features that are difficult to meet the standard Gaussian innovation assumption. Cont [25] documents a set of stylized facts for asset returns, including heavy tails, volatility clustering, and extreme fluctuations. These features imply that Gaussian models tend to put insufficient probability mass on large shocks, which can lead to poorly predictive distributions and systematic underestimation of tail risk. This concern is reinforced by the stochastic volatility work. For example, Kim et al. [26] shows that models with latent time-varying volatility provide a richer account of financial dynamics than specifications based on constant Gaussian innovations. More generally, this work suggests that Gaussian VAR models may be useful for describing conditional mean dynamics, but are often inadequate for

modeling distributional asymmetry, time-varying uncertainty, and tail-risk behavior.

These limitations have motivated a broad work on more flexible innovation specifications in Bayesian time-series models. One common approach is to replace the Gaussian distribution with a heavy-tailed alternative such as the Student- t distribution, which is better suited to capturing large shocks [27, 28]. A second approach is to introduce latent stochastic volatility processes, allowing the innovation variance to evolve and accommodating the volatility clustering often observed in financial data [29–31]. A third type of work uses mixture-based specifications to capture heterogeneous shock regimes and more general non-Gaussian distributional features [32].

For these reasons, relaxing the Gaussian innovation assumption is a natural extension of Bayesian VAR models in financial forecasting. The next subsection reviews several flexible innovation specifications to improve predictive density estimation and tail-risk assessment.

Flexible Innovation Models

To address the limitations of Gaussian innovations, the literature has developed a range of more flexible specifications for Bayesian time-series models [33, 34]. Although these models differ in form, they share a common goal: to provide a richer description of the predictive distribution by accommodating features such as heavy tails, regime heterogeneity, and time-varying volatility. In the context of Bayesian VARs, three particularly important approaches are Student- t innovations, mixture-based innovations, and stochastic volatility models [35, 36].

Student- t Innovations

A simple way to relax the Gaussian innovation assumption is to let the shocks follow a Student- t distribution. In a VAR setting, this can be written as

$$\varepsilon_t \sim t_\nu(0, \Sigma),$$

where ν denotes the degrees of freedom and Σ is the scale matrix. Relative to the Gaussian case, the Student- t distribution places more probability mass in the tails, so it is better able to accommodate unusually large shocks. This feature is particularly useful in financial and macroeconomic applications, where crises and abrupt regime changes can strongly influence forecasting performance.

An additional advantage is that the Student- t specification remains computationally convenient in Bayesian analysis. A common representation is

$$\varepsilon_t \mid \lambda_t \sim \mathcal{N}(0, \Sigma/\lambda_t), \quad \lambda_t \sim \text{Gamma}(\nu/2, \nu/2),$$

so that the model can be interpreted as a Gaussian system with an observation-specific latent scale. This formulation preserves much of the tractability of Gaussian VARs while providing greater robustness to outliers and heavy-tailed behavior.

Recent work shows that this idea remains highly relevant in modern forecasting problems. For example, Anttonen et al. [37] consider SVAR models with skewed generalized- t errors, showing that non-Gaussian innovations can provide a more flexible description of macroeconomic shocks. More recently, Ertl et al. [38] study inflation forecasting in turbulent times using a mixed-frequency BVAR and find that Student- t innovations offer a natural way to capture tail events that became important after the pandemic and the energy-price shock period. These studies suggest that Student- t -type innovations provide a useful middle ground between the simplicity of Gaussian models and the need for a more realistic treatment of rare but influential shocks. For this reason, they are often a natural first step when extending Bayesian VARs beyond the Gaussian benchmark.

Mixture-of-Normals Innovations

A second way to relax the Gaussian assumption is to model the innovations as a mixture of Gaussian distributions. In its simplest form, this can be written as

$$\varepsilon_t \sim \sum_{k=1}^K \pi_k \mathcal{N}(0, \Sigma_k), \quad \sum_{k=1}^K \pi_k = 1,$$

where each component represents a different shock regime and the mixing weights π_k determine how likely each regime is. Equivalently, we can introduce a latent state variable z_t such that

$$z_t \sim \text{Categorical}(\pi_1, \dots, \pi_K), \quad \varepsilon_t \mid z_t = k \sim \mathcal{N}(0, \Sigma_k).$$

This representation is intuitive and flexible: instead of forcing all shocks to follow a single distribution, the model allows different observations to arise from different latent states.

The appeal of this approach is that it can capture features that are difficult to describe with a single Gaussian or even a single heavy-tailed distribution. In particular, mixture innovations can accommodate multimodality, asymmetry, and heterogeneous shock behavior, which are often present in financial time series. They are especially useful when the data appear to switch between calm and turbulent periods or when extreme observations are concentrated in only part of the sample.

A representative contribution is Chiu et al. [39], who model VAR disturbances using a finite mixture of normal distributions with regime switching. Their framework shows that mixture-based shocks can generate substantial forecasting gains relative to more standard specifications, especially when the data exhibit pronounced non-Gaussian behavior. More recent work has pushed this idea further in large and flexible Bayesian systems. For example, Huber and Koop [40] develop Bayesian VARs with nonparametric shocks based on a Dirichlet process mixture, allowing the innovation distribution to adapt more freely while maintaining computationally efficient inference. In a related direction, Gretener et al. [41] propose a dynamic mixture VAR in which the mixture weights evolve according to the predictive likelihood score, allowing the model to respond more directly to changing market conditions. These studies show that mixture innovations provide a flexible way to model heterogeneous shock regimes and complex distributional shapes in multivariate time-series systems [42, 43]. For this reason, they form an important class of alternatives to both Gaussian and single heavy-tailed innovation specifications [44, 45].

Stochastic Volatility Models

A third important extension is to allow the innovation variance to evolve through a latent stochastic volatility process. In a Bayesian VAR, this idea is represented as

$$\varepsilon_t \sim \mathcal{N}(0, \Sigma_t),$$

where the covariance matrix Σ_t changes over time rather than remaining fixed. In a simple diagonal specification, we can write

$$\log h_{i,t} = \mu_i + \phi_i(\log h_{i,t-1} - \mu_i) + \eta_{i,t},$$

so that each variance component $h_{i,t}$ follows a persistent latent process. This formulation is attractive because it directly captures volatility clustering, a pervasive feature of financial time series in which periods of calm and turbulence tend to persist.

The main benefit of stochastic volatility models is that they allow uncertainty itself to be dynamic. This is particularly important in financial forecasting, where shock magnitudes often vary over time and constant-variance Gaussian models may provide a misleading description of predictive uncertainty. In this sense, stochastic volatility models enrich Bayesian VARs not by changing the conditional mean, but by giving a more realistic description of the full predictive distribution [46, 47].

A representative contribution is Carriero et al. [48], who develop large Bayesian VARs with stochastic volatility and non-conjugate priors. Their work shows that stochastic volatility can be incorporated into high-dimensional systems while remaining useful for forecasting and structural analysis. In a related direction, Chan et al. [49] introduce large Bayesian VARs with factor stochastic volatility, showing that this structure can help address ordering issues while remaining suitable for large multivariate systems. More recently, Cross et al. [50] extend the framework further by allowing stochastic volatility to enter the conditional mean. This can link macro-financial uncertainty directly to the evolution of the observed variables. These studies show that stochastic volatility models are valuable when the dynamics of uncertainty are central to the forecasting problem rather than merely a nuisance feature of the data. For this reason, stochastic volatility has become one of the most important alternatives to constant-variance Gaussian innovations in modern Bayesian VAR analysis [51, 52].

Forecast Evaluation and Tail-Risk Metrics

Because our goal is to compare predictive distributions rather than only point forecasts, forecast evaluation must focus on how well a model describes the full distribution of future outcomes. A natural starting point is the predictive density $p_t(y_t)$ itself. One widely used measure is the log predictive score,

$$\text{LPS}_t = \log p_t(y_t),$$

that assign high probability to the realized outcome. Beyond scalar scores, density forecasts are assessed using calibration diagnostics based on the probability integral transform (PIT), which provides a direct means of assessing whether the forecast distribution is consistent with the observed data [53].

When the main interest lies in specific parts of the predictive distribution, global scoring rules may be complemented by tail-focused comparisons. As a result, Amisano and Giacomini [54] propose weighted likelihood methods for comparing density forecasts, allowing the evaluation to place greater emphasis on economically important regions such as the left tail. This is very relevant in financial forecasting, where models may have similar overall fit but differ in how well they capture downside risk.

Recent statistical work has clarified how these objects should be evaluated. In particular, Fissler and Ziegel [55] show that VaR and ES are jointly available, which provides a theoretical foundation for constructing consistent scoring functions for VaR and ES. Building on this perspective, Nolde and Ziegel [56] discuss the roles in risk evaluation, while Bayer and Dimitriadis [57] develop regression backtests for Expected Shortfall. More recently, Hoga and Demetrescu [58] propose monitoring procedures for VaR and ES forecasts, highlighting that risk evaluation should not be viewed as a one-time exercise but as an ongoing component of model assessment.

3 Theoretical Framework

In this section, we present the Bayesian formulation of the models considered in this work, including Gaussian VAR, Student- t VAR, mixture VAR, and stochastic volatility VAR. We also introduce the posterior predictive framework and risk measures used for evaluation.

Bayesian VAR

Consider a VAR(p) model

$$y_t = Bx_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \Sigma), \quad (1)$$

where $x_t = (1, y_{t-1}, \dots, y_{t-p})^\top$ is a stacked regressor.

Let $Y = XB^\top + E$. The likelihood is

$$p(Y | B, \Sigma) \propto |\Sigma|^{-n/2} \exp \left(-\frac{1}{2} \text{tr}[(Y - XB^\top)^\top (Y - XB^\top) \Sigma^{-1}] \right). \quad (2)$$

We impose the conjugate prior

$$B | \Sigma \sim MN(B_0, \Sigma, V_0), \quad \Sigma \sim IW(S_0, \nu_0). \quad (3)$$

The posterior takes the form

$$B | \Sigma, Y \sim MN(B_n, \Sigma, V_n), \quad \Sigma | Y \sim IW(S_n, \nu_0 + n), \quad (4)$$

where $V_n = (V_0^{-1} + X^\top X)^{-1}$.

The posterior predictive distribution is a multivariate Student- t :

$$y_{T+1} | Y \sim t(B_n x_{T+1}, S_n). \quad (5)$$

Student- t VAR

We extend the model by assuming heavy-tailed innovations

$$\varepsilon_t \sim t_\nu(0, \Sigma). \quad (6)$$

Using the scale-mixture representation,

$$\varepsilon_t | \lambda_t \sim \mathcal{N}(0, \Sigma/\lambda_t), \quad \lambda_t \sim \text{Gamma}(\nu/2, \nu/2). \quad (7)$$

Conditional on Λ , the model reduces to a weighted Gaussian VAR. Posterior inference is conducted via Gibbs sampling by iterating over

$$\lambda_t, \quad B | \Sigma, \Lambda, Y, \quad \Sigma | B, \Lambda, Y. \quad (8)$$

Mixture VAR

We model innovations as a finite mixture:

$$\varepsilon_t \sim \sum_{k=1}^K \pi_k \mathcal{N}(0, \Sigma_k). \quad (9)$$

Introduce latent states z_t , with

$$y_t | z_t = k \sim \mathcal{N}(Bx_t, \Sigma_k). \quad (10)$$

Posterior inference proceeds via Gibbs sampling over

$$z_t, \quad \pi, \quad \Sigma_k, \quad B. \quad (11)$$

Stochastic Volatility VAR

We allow time-varying covariance

$$\varepsilon_t \sim \mathcal{N}(0, D_t), \quad D_t = \text{diag}(e^{h_t}). \quad (12)$$

The latent log-volatility follows

$$h_t = \mu + \phi(h_{t-1} - \mu) + \eta_t. \quad (13)$$

Posterior inference combines Gaussian regression updates for B with sampling of latent volatilities.

Posterior Predictive and Risk Measures

The predictive distribution is

$$p(y_{T+1} | Y) = \int p(y_{T+1} | \theta) p(\theta | Y) d\theta. \quad (14)$$

We evaluate models using:

Log predictive score:

$$\text{LPS} = \log p(y_{T+1}^{\text{obs}} | Y). \quad (15)$$

Value-at-Risk:

$$\text{VaR}_\alpha = \inf\{r : \Pr(R_{T+1} \leq r) \geq \alpha\}. \quad (16)$$

Expected shortfall:

$$\text{ES}_\alpha = \mathbb{E}[R_{T+1} | R_{T+1} \leq \text{VaR}_\alpha]. \quad (17)$$

4 Data Source and Preprocessing

Data Description

The dataset consists of daily financial returns for four major U.S. equity exchange-traded funds (ETFs): SPY (S&P 500), QQQ (NASDAQ 100), DIA (Dow Jones Industrial Average), and IWN (Russell 2000 Value). These assets provide broad coverage of the U.S. equity market across different sectors and capitalization profiles.

The sample spans from January 2002 to February 2026, yielding over 6,000 observations per series. This period includes multiple market regimes, including the 2008 financial crisis and the COVID-19 shock, making it well-suited for studying heavy tails and volatility dynamics.

Data Collection

Price data were obtained using a standardized data pipeline implemented in Python. The data were aligned across assets to ensure a consistent set of trading days. Any missing values were handled by taking the intersection of available dates, resulting in a fully balanced panel. The resulting dataset is a multivariate time series with synchronized observations across all four assets.

Return Construction

Daily log returns were computed from adjusted closing prices. For each asset, returns were defined as

$$r_t = \log P_t - \log P_{t-1},$$

where P_t denotes the adjusted closing price at time t . Daily log returns for the four ETFs are shown in (Fig. A1).

This transformation stabilizes variance and ensures additivity over time, which is standard in financial econometrics and facilitates modeling within the VAR framework.

Standardization

To improve numerical stability and comparability across series, each return series was standardized to have approximately zero mean and unit variance. This step is particularly important for Bayesian estimation and MCMC sampling, as it prevents scale imbalances from affecting convergence and inference. Standardization does not alter the dependence structure of the data but simplifies interpretation of model parameters and diagnostics.

Exploratory Data Analysis

Initial exploratory analysis reveals several well-known stylized facts of financial returns:

- **Heavy tails:** Empirical distributions exhibit extreme observations far beyond what would be expected under a Gaussian assumption, as confirmed by high kurtosis values (Fig. A2).
- **Volatility clustering:** Periods of high volatility tend to cluster together, indicating time-varying variance (Fig. A3).
- **Strong cross-asset correlation:** The ETF returns display high contemporaneous correlation, justifying a multivariate modeling approach (Fig. A4).
- **Near-zero mean:** Returns are centered around zero, consistent with efficient market behavior.

These features motivate the use of more flexible probabilistic models beyond the standard Gaussian VAR, particularly those capable of capturing heavy tails, regime changes, and time-varying volatility.

5 Methodology

In this section, we describe the empirical implementation of the models, including estimation procedures, forecasting design, and evaluation methods. We consider both point estimation and Bayesian approaches under a unified framework, allowing for a direct comparison of their predictive performance. We estimate a multivariate VAR using all four ETF return series. The analysis focuses on SPY for evaluation, as it serves as a proxy for overall market dynamics. While the VAR framework jointly models all assets, reporting results for a single representative series avoids redundancy without loss of generality.

General Framework

We consider a multivariate time series $\{y_t\}_{t=1}^T$, where $y_t \in \mathbb{R}^m$ represents asset returns. All models are based on a VAR(p) specification

$$y_t = Bx_t + \varepsilon_t, \quad (18)$$

where $x_t = (1, y_{t-1}, \dots, y_{t-p})^\top$.

The models differ in the specification of the innovation term ε_t , while sharing the same regression structure.

To evaluate predictive performance, we adopt a rolling-window forecasting scheme. Let W denote the window size. At each time t , parameters are estimated using observations $t-W, \dots, t-1$, and a one-step-ahead prediction for y_t is generated. This process is repeated across the out-of-sample period.

Point Estimation Approach

We first implement point estimation methods as computationally efficient benchmarks.

Gaussian VAR

Under Gaussian innovations, the model reduces to a standard multivariate linear regression. Writing

$$Y = XB^\top + E, \quad (19)$$

the ordinary least squares estimator is

$$\hat{B} = (X^\top X)^{-1} X^\top Y. \quad (20)$$

The covariance matrix is estimated using residuals:

$$\hat{\Sigma} = \frac{1}{n} (Y - X\hat{B}^\top)^\top (Y - X\hat{B}^\top). \quad (21)$$

Given $(\hat{B}, \hat{\Sigma})$, predictive distributions are constructed as

$$y_{t+1} \sim \mathcal{N}(\hat{B}x_{t+1}, \hat{\Sigma}). \quad (22)$$

Student-t VAR

For the Student- t model, we use a quasi-maximum likelihood approach based on its scale-mixture representation:

$$y_t \mid \lambda_t \sim \mathcal{N}(Bx_t, \Sigma/\lambda_t). \quad (23)$$

Estimation proceeds iteratively by alternating between:

- updating latent scale variables λ_t ,
- updating B and Σ via weighted least squares.

At each iteration, the weighted regression takes the form

$$\hat{B} = (X^\top \Lambda X)^{-1} X^\top \Lambda Y, \quad (24)$$

where $\Lambda = \text{diag}(\lambda_t)$.

In practice, the degrees-of-freedom parameter ν is fixed, and convergence is achieved after a small number of iterations.

Mixture VAR

For the mixture VAR, we model innovations as

$$\varepsilon_t \sim \sum_{k=1}^K \pi_k \mathcal{N}(0, \Sigma_k). \quad (25)$$

We estimate parameters using an Expectation-Maximization (EM) algorithm. In the E-step, posterior regime probabilities are computed:

$$r_{tk} = \frac{\pi_k \phi(y_t; Bx_t, \Sigma_k)}{\sum_{j=1}^K \pi_j \phi(y_t; Bx_t, \Sigma_j)}. \quad (26)$$

In the M-step, parameters are updated using weighted regressions:

$$\hat{B} = \left(\sum_t r_{tk} x_t x_t^\top \right)^{-1} \left(\sum_t r_{tk} x_t y_t \right). \quad (27)$$

We fix the number of components to $K = 2$, corresponding to low- and high-volatility regimes.

Bayesian Estimation

We now describe the Bayesian estimation procedure for the models. In contrast to point estimation, which produces a single estimate of the parameters, the Bayesian approach characterizes the full posterior distribution and naturally incorporates parameter uncertainty into predictive inference.

Let θ denote all model parameters and latent variables. Bayesian inference is based on the posterior distribution

$$p(\theta | Y) \propto p(Y | \theta) p(\theta), \quad (28)$$

which combines the likelihood and prior distribution.

For all models considered, posterior inference is carried out using Markov Chain Monte Carlo (MCMC), specifically Gibbs sampling. This approach exploits conditional conjugacy to sample sequentially from the full conditional distributions of each parameter block.

MCMC Algorithm

Posterior inference is conducted via Gibbs sampling, exploiting conditional conjugacy and latent-variable augmentation for non-Gaussian models. Let θ denote all model parameters and latent variables. The goal is to sample from the posterior

$$p(\theta | Y).$$

At each iteration $s = 1, \dots, S$, we sequentially draw from full conditional distributions.

Gaussian VAR : Under conjugate priors, the full conditionals are available in closed form:

$$B | \Sigma, Y \sim MN(B_n, \Sigma, V_n), \quad (29)$$

$$\Sigma | B, Y \sim IW(S_n, \nu_n), \quad (30)$$

where

$$V_n = (V_0^{-1} + X^\top X)^{-1}, \quad B_n = V_n(V_0^{-1} B_0 + X^\top Y).$$

Student- t VAR : Using the scale-mixture representation, introduce latent variables λ_t such that

$$y_t \mid \lambda_t \sim \mathcal{N}(Bx_t, \Sigma/\lambda_t), \quad \lambda_t \sim \text{Gamma}(\nu/2, \nu/2).$$

The Gibbs updates are:

$$\lambda_t \mid y_t, B, \Sigma \sim \text{Gamma}\left(\frac{\nu + m}{2}, \frac{\nu + \delta_t}{2}\right), \quad (31)$$

$$\delta_t = (y_t - Bx_t)^\top \Sigma^{-1} (y_t - Bx_t), \quad (32)$$

Define $\Lambda = \text{diag}(\lambda_t)$. Then

$$B \mid \Sigma, \Lambda, Y \sim MN(B_n, \Sigma, V_n), \quad (33)$$

$$\Sigma \mid B, \Lambda, Y \sim IW(S_n, \nu_n), \quad (34)$$

with weighted sufficient statistics:

$$X^\top \Lambda X, \quad X^\top \Lambda Y.$$

Mixture VAR : Introduce latent regime indicators $z_t \in \{1, \dots, K\}$:

$$y_t \mid z_t = k \sim \mathcal{N}(Bx_t, \Sigma_k).$$

The full conditional updates are:

$$\Pr(z_t = k \mid \cdot) \propto \pi_k \mathcal{N}(y_t; Bx_t, \Sigma_k), \quad (35)$$

$$\pi \mid Z \sim \text{Dirichlet}(\alpha_1 + n_1, \dots, \alpha_K + n_K), \quad (36)$$

where $n_k = \sum_t \mathbf{1}(z_t = k)$. Conditional on Z , each regime is a Gaussian regression:

$$B \mid \{\Sigma_k\}, Z, Y \sim MN(B_n, \Sigma^*, V_n), \quad (37)$$

$$\Sigma_k \mid B, Z, Y \sim IW(S_k, \nu_k), \quad (38)$$

where sufficient statistics are computed using observations assigned to regime k .

Stochastic Volatility VAR We assume

$$y_t \mid h_t \sim \mathcal{N}(Bx_t, D_t), \quad D_t = \text{diag}(e^{h_t}).$$

The latent log-volatility follows

$$h_t = \mu + \phi(h_{t-1} - \mu) + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma_h^2).$$

The Gibbs sampler alternates between:

$$B \mid h, Y \sim MN(B_n, \Sigma_h, V_n), \quad (39)$$

$$h_t \mid \cdot \sim p(h_t \mid Y, B, \theta_h), \quad (40)$$

$$\theta_h \mid h \sim p(\theta_h \mid h), \quad (41)$$

where sampling of h_t is performed using Metropolis-Hastings or forward-filtering backward-sampling due to nonlinearity.

Posterior Predictive Sampling Given posterior draws $\{\theta^{(s)}\}_{s=1}^S$, predictive samples are generated as

$$y_{T+1}^{(s)} \sim p(y_{T+1} \mid \theta^{(s)}),$$

yielding the Monte Carlo approximation

$$p(y_{T+1} \mid Y) \approx \frac{1}{S} \sum_{s=1}^S p(y_{T+1} \mid \theta^{(s)}).$$

Convergence Diagnostics Convergence is assessed using effective sample size (ESS), trace plots, and stability of posterior summaries across chains. Burn-in and thinning are applied to reduce autocorrelation and ensure reliable posterior inference.

Gaussian VAR

For the Gaussian VAR with conjugate priors, the posterior distributions have closed-form expressions. At each iteration s , we sample:

$$B^{(s)} \sim p(B \mid \Sigma^{(s-1)}, Y), \quad (42)$$

$$\Sigma^{(s)} \sim p(\Sigma \mid B^{(s)}, Y). \quad (43)$$

Both conditional distributions follow standard matrix-normal and inverse-Wishart forms. This results in an efficient Gibbs sampler with straightforward implementation.

Student- t VAR

For the Student- t VAR, we introduce latent scale variables λ_t , which convert the model into a conditionally Gaussian system:

$$y_t \mid \lambda_t \sim \mathcal{N}(Bx_t, \Sigma/\lambda_t). \quad (44)$$

The Gibbs sampler alternates between:

$$\lambda_t^{(s)} \sim p(\lambda_t \mid y_t, B^{(s-1)}, \Sigma^{(s-1)}), \quad (45)$$

$$B^{(s)} \sim p(B \mid \Sigma^{(s-1)}, \Lambda^{(s)}, Y), \quad (46)$$

$$\Sigma^{(s)} \sim p(\Sigma \mid B^{(s)}, \Lambda^{(s)}, Y), \quad (47)$$

where $\Lambda = \text{diag}(\lambda_t)$.

This augmentation allows us to retain conjugacy while capturing heavy-tailed behavior through observation-specific scaling.

Mixture VAR

For the mixture VAR, we introduce latent regime indicators z_t , where

$$y_t \mid z_t = k \sim \mathcal{N}(Bx_t, \Sigma_k). \quad (48)$$

The Gibbs sampler iterates over:

$$z_t^{(s)} \sim p(z_t \mid y_t, B^{(s-1)}, \Sigma^{(s-1)}, \pi^{(s-1)}), \quad (49)$$

$$\pi^{(s)} \sim p(\pi \mid Z^{(s)}), \quad (50)$$

$$\Sigma_k^{(s)} \sim p(\Sigma_k \mid B^{(s-1)}, Z^{(s)}, Y), \quad (51)$$

$$B^{(s)} \sim p(B \mid Z^{(s)}, \Sigma^{(s)}, Y). \quad (52)$$

This framework allows the model to capture regime-dependent dynamics, with different covariance structures across latent states.

Stochastic Volatility VAR

For the stochastic volatility VAR, the covariance matrix evolves over time through latent log-volatility processes h_t . Conditional on h_t , the model is Gaussian:

$$y_t \mid h_t \sim \mathcal{N}(Bx_t, D_t), \quad (53)$$

where $D_t = \text{diag}(e^{h_t})$.

The Gibbs sampler combines:

$$B^{(s)} \sim p(B \mid h^{(s-1)}, Y), \quad (54)$$

$$h_t^{(s)} \sim p(h_t \mid Y, B^{(s)}, \theta_h^{(s-1)}), \quad (55)$$

$$\theta_h^{(s)} \sim p(\theta_h \mid h^{(s)}), \quad (56)$$

where θ_h denotes volatility parameters.

Sampling of h_t typically requires a Metropolis-Hastings step due to nonlinearity.

Posterior Predictive Distribution

Given posterior samples $\{\theta^{(s)}\}_{s=1}^S$, the predictive distribution is approximated as

$$p(y_{t+1} \mid Y) \approx \frac{1}{S} \sum_{s=1}^S p(y_{t+1} \mid \theta^{(s)}). \quad (57)$$

This Monte Carlo approximation captures both model uncertainty and parameter uncertainty, leading to smoother and more robust predictive densities.

Compared to point estimation, the Bayesian approach provides a richer characterization of uncertainty, which is particularly important for tail-risk estimation.

Forecasting Procedure

At each forecast origin, we construct the predictive distribution of y_{t+1} .

For point estimation:

$$y_{t+1} \sim p(y_{t+1} \mid \hat{\theta}). \quad (58)$$

For Bayesian inference:

$$p(y_{t+1} \mid Y) \approx \frac{1}{S} \sum_{s=1}^S p(y_{t+1} \mid \theta^{(s)}). \quad (59)$$

In both cases, we generate simulated draws of future returns, which are used to approximate predictive densities.

6 Evaluation Metrics

We evaluate predictive performance using probabilistic scoring rules and tail-risk measures.

Log Predictive Score

$$\text{LPS} = \sum_t \log p(y_t | y_{1:t-1}),$$

which measures predictive density accuracy [59, 60].

Value-at-Risk

$$\text{VaR}_\alpha = \inf\{x : P(r_{t+1} \leq x) \geq \alpha\}.$$

Expected Shortfall

$$\text{ES}_\alpha = \mathbb{E}[r_{t+1} \mid r_{t+1} \leq \text{VaR}_\alpha].$$

These metrics evaluate both predictive density quality and tail-risk calibration [61, 62].

7 Empirical Results

In-Sample Results

Parameter Estimates

Across all models, estimated VAR coefficients are relatively small in magnitude, indicating limited linear predictability in daily returns. This is consistent with the weak-form efficiency hypothesis in financial markets. The dependence structure is instead primarily captured through the covariance matrices.

The Gaussian and Student- t models produce broadly similar coefficient estimates, suggesting that heavy-tailed modeling affects uncertainty and residual structure more than mean dynamics. However, covariance estimates differ more substantially across models. In particular, the Student- t model yields slightly lower marginal variances, reflecting its ability to absorb extreme observations through its latent scaling mechanism rather than inflating variance.

Table 1: Estimated VAR coefficients for Gaussian and Student- t models

	intercept	SPY _{$t-1$}	QQQ _{$t-1$}	DIA _{$t-1$}	IWN _{$t-1$}	SPY _{$t-2$}	QQQ _{$t-2$}	DIA _{$t-2$}	IWN _{$t-2$}
Gaussian VAR									
SPY	-0.0002	-0.1030	0.0746	-0.1085	0.0625	-0.1586	0.0441	0.0974	0.0197
QQQ	-0.0004	-0.0576	0.0472	-0.1151	0.0401	-0.0592	-0.0137	0.0345	0.0260
DIA	-0.0002	-0.0455	0.0641	-0.1682	0.0577	-0.1398	0.0329	0.0907	0.0242
IWN	-0.0002	-0.0217	0.0273	-0.0900	-0.0044	-0.0842	0.0120	0.1193	-0.0230
Student-t VAR (Posterior Mean)									
SPY	0.0770	-0.0363	0.0254	-0.0690	0.0160	0.0060	-0.0062	0.0182	-0.0193
QQQ	0.0994	0.0167	-0.0010	-0.0860	0.0146	0.0467	-0.0279	-0.0146	-0.0112
DIA	0.0574	0.0350	0.0078	-0.1207	0.0103	0.0150	-0.0059	0.0126	-0.0166
IWN	0.0671	0.0497	0.0006	-0.0900	-0.0212	0.0745	-0.0227	0.0220	-0.0584

The mixture model further distinguishes between regimes, with one component exhibiting low variance (normal conditions) and the other showing dramatically larger covariance values, corresponding to stress periods. This separation highlights the presence of distinct volatility regimes in the data (Fig. A5).

Table 2: Estimated covariance matrices across models

	SPY	QQQ	DIA	IWN
Gaussian VAR				
SPY	0.9893	0.8766	0.9268	0.8297
QQQ	0.8766	0.9901	0.8258	0.7315
DIA	0.9268	0.8258	0.9862	0.8403
IWN	0.8297	0.7315	0.8403	0.9904
Student-<i>t</i> VAR (Posterior Mean)				
SPY	0.7162	0.8152	0.6480	0.8030
QQQ	0.8152	1.1356	0.7012	0.8698
DIA	0.6480	0.7012	0.6472	0.7446
IWN	0.8030	0.8698	0.7446	1.2613

Residual Diagnostics

Residual diagnostics reveal clear departures from Gaussian assumptions. While residuals from the Gaussian VAR are centered around zero, they exhibit substantial excess kurtosis, indicating the presence of heavy tails. Additionally, residual time series plots show persistent volatility clustering, suggesting that constant-variance assumptions are inadequate.

Cross-asset residual correlations remain high, reflecting strong co-movement in shocks across markets. These findings confirm that the Gaussian VAR fails to fully capture key features of the data.

In contrast, the Student-*t* VAR produces standardized residuals with significantly reduced kurtosis, indicating that heavy-tailed behavior has been effectively modeled. The mixture model further accounts for extreme observations through regime switching, while the stochastic volatility model captures time-varying variance but does not fully eliminate tail risk (Fig. A6).

Distributional Fit

Comparing predictive distributions across models highlights important differences in how each specification captures uncertainty. The Gaussian VAR produces relatively thin-tailed predictive densities, which tend to underestimate the probability of extreme events (Fig. A7).

The Student-*t* VAR improves the fit by introducing heavy tails through latent scale parameters. These latent variables dynamically adjust the effective variance of observations, allowing the model to downweight extreme shocks. This is reflected in near-Gaussian behavior of standardized residuals (Fig. A8).

The mixture VAR captures non-Gaussianity through discrete regime switching. One regime corresponds to typical market conditions, while the other represents high-volatility periods with substantially larger covariance. The model assigns relatively small probability to the high-volatility regime, but these periods align with known market stress events.

Finally, the stochastic volatility VAR models continuous variation in volatility over time. The estimated latent volatility processes closely track absolute returns, capturing volatility clustering. However, compared to the mixture model, the SV specification produces less extreme tail behavior in predictive distributions (Fig. A9).

Point Estimation Results

We now examine the in-sample behavior of the models under point estimation. While the overall coefficient patterns are broadly consistent with the Bayesian estimates, the main differences arise in the residual structure and distributional fit.

Residual Diagnostics

Figure A10 compares residual distributions across models. The Gaussian VAR residuals exhibit clear excess kurtosis, indicating that the normality assumption fails to capture extreme observations. In contrast, the Student- t model produces more concentrated residuals, reflecting its ability to accommodate heavy tails. The mixture model further improves the fit by capturing regime-dependent variability.

These findings are consistent with the Bayesian results, confirming that Gaussian innovations fail to capture heavy-tailed behavior, while Student- t and mixture models provide more robust in-sample fits.

Distributional Fit

Figure A11 shows the predictive distributions implied by point estimation. The Gaussian model produces relatively thin-tailed densities, whereas the Student- t model exhibits heavier tails. The mixture model generates more flexible shapes, reflecting its ability to represent multiple volatility regimes.

Overall, the point estimation results lead to similar qualitative conclusions as the Bayesian analysis: non-Gaussian specifications improve the in-sample representation of financial returns, particularly in capturing heavy tails and regime-dependent variability.

Out-of-Sample Results

Rolling Forecast Setup

Out-of-sample performance is evaluated using a rolling window forecasting scheme. At each step, models are re-estimated on a fixed-size training window and used to generate one-step-ahead predictive distributions. This procedure mimics a real-time forecasting setting and allows for consistent comparison across models.

For each forecast origin, predictive distributions are obtained via simulation, from which summary statistics and risk measures are computed. The evaluation focuses on a single target series (SPY), though the framework is fully multivariate. Forecast accuracy is assessed using the log predictive score (LPS), while risk is evaluated using Value-at-Risk (VaR) and Expected Shortfall (ES) at the 5% level.

Predictive Performance Comparison

Table results show clear differences in predictive performance across models. The mixture VAR achieves the highest (least negative) mean log predictive score, indicating superior ability to assign probability mass to realized outcomes. This improvement is driven by its capacity to adapt to regime changes and capture extreme events.

The Student- t VAR performs second best, benefiting from heavy-tailed innovations that improve robustness to outliers. In contrast, the Gaussian VAR underperforms due to its inability to cap-

ture tail behavior, while the stochastic volatility VAR, although capturing time-varying variance, produces less accurate predictive densities.

Risk measures further highlight these differences. The mixture model produces the most conservative expected shortfall, reflecting strong sensitivity to tail risk. The Student- t model provides moderate tail estimates, while the Gaussian and SV models tend to underestimate extreme outcomes.

Table 3: Out-of-sample predictive performance across models

Model	n	Mean LPS	VaR _{5%}	ES _{5%}	Avg. Pred. Std
Mixture	101	-1.9656	-1.7524	-2.8140	1.1608
Student- t	101	-2.1010	-1.5624	-2.1849	0.9537
Gaussian	101	-2.1400	-1.8480	-2.3369	1.1775
SV	101	-2.1809	-1.3686	-1.7726	0.9531

Time-Varying Performance

To better understand model performance over time, we analyze rolling and cumulative log predictive scores. All models exhibit sharp declines in predictive performance during periods of market stress, reflecting the difficulty of forecasting extreme events.

However, differences across models become more pronounced in these periods. The mixture VAR shows greater resilience, with smaller performance drops and faster recovery following shocks. The Student- t model also performs relatively well, while the Gaussian and SV models experience more severe deterioration.

Cumulative log predictive scores further reinforce these findings, with the mixture model consistently outperforming alternatives over the evaluation horizon. Rolling averages of LPS highlight short-term fluctuations and show that flexible models provide more stable performance during volatile periods (Fig. A12).

Point Estimation Results

We next evaluate out-of-sample predictive performance under point estimation. The same rolling-window forecasting scheme is used, ensuring direct comparability with the Bayesian results.

Predictive Performance

Across models, predictive accuracy differs substantially depending on the innovation specification. The Gaussian VAR tends to assign insufficient probability mass to extreme observations, leading to lower predictive scores. The Student- t model improves robustness by incorporating heavy tails, while the mixture model achieves the best overall performance by adapting to regime changes.

These patterns are consistent with the Bayesian results, although the absence of parameter uncertainty in point estimation leads to slightly less flexible predictive densities.

Risk Evaluation

Figure A13 compares VaR violations across models. The Gaussian model exhibits more frequent violations, indicating systematic underestimation of tail risk. The Student- t model reduces violations by accounting for heavy tails, while the mixture model provides the most stable performance, reflecting its ability to capture extreme market conditions.

Comparison with Bayesian Results

While both approaches yield consistent rankings across models, Bayesian inference generally produces smoother predictive distributions and more stable risk estimates due to the incorporation of parameter uncertainty. Nevertheless, the point estimation results confirm the same central conclusion: models with flexible, non-Gaussian innovations significantly outperform the Gaussian benchmark in both predictive accuracy and tail-risk estimation.

Model Comparison(Full Bayesian Estimation)

Predictive Accuracy

Across all evaluation metrics, the mixture VAR achieves the strongest predictive performance. It consistently attains the highest log predictive score both in-sample and out-of-sample, indicating superior ability to assign probability mass to realized outcomes. This improvement is particularly pronounced during periods of market stress, where regime-switching allows the model to adapt to sudden changes in volatility.

The Student- t VAR provides the second-best performance, demonstrating that heavy-tailed innovations alone significantly improve predictive accuracy relative to the Gaussian benchmark. By contrast, the Gaussian VAR underperforms due to its inability to capture extreme observations, while the stochastic volatility VAR, although capturing time-varying variance, produces less accurate predictive densities overall.

These results suggest that explicitly modeling non-Gaussian features of the data is critical for accurate probabilistic forecasting.

Risk Estimation

The models differ substantially in their characterization of tail risk. The Gaussian VAR produces the least extreme Value-at-Risk and Expected Shortfall estimates, reflecting its thin-tailed assumption and resulting underestimation of extreme events.

The Student- t VAR improves upon this by incorporating heavy tails, yielding more realistic tail risk measures. The mixture VAR produces the most conservative Expected Shortfall estimates, reflecting its ability to capture rare but severe high-volatility regimes. This leads to stronger tail sensitivity and improved modeling of downside risk.

The stochastic volatility VAR captures time-varying risk levels but does not generate tail behavior as extreme as the mixture model, resulting in comparatively moderate risk estimates.

Overall, models that explicitly incorporate non-Gaussian structure provide more reliable measures of financial risk (Fig. [A14](#)).

Interpretability

Each model presents a different trade-off between interpretability and flexibility. The Gaussian VAR is the most interpretable, with straightforward linear dynamics and a constant covariance structure. However, this simplicity comes at the cost of poor fit to real-world data.

The Student- t VAR retains much of the interpretability of the Gaussian model while introducing latent scale parameters that account for extreme observations. This provides a relatively simple and effective extension.

The mixture VAR offers a highly interpretable regime-based framework, where each component corresponds to a distinct market state (e.g., normal versus crisis periods). While more complex, the regime interpretation provides meaningful economic insight.

The stochastic volatility VAR models continuous changes in volatility over time, offering a realistic representation of financial markets. However, the latent volatility processes are less directly interpretable compared to discrete regimes.

In summary, there is a clear trade-off between model complexity and interpretability, with the mixture and stochastic volatility models offering greater flexibility at the cost of increased computational and conceptual complexity.

8 Conclusion

Summary of Findings

This paper investigates the impact of non-Gaussian innovations in vector autoregression models for financial time series, under both Bayesian and point estimation frameworks. Starting from a Gaussian VAR baseline, we progressively introduce more flexible specifications, including Student- t innovations, mixture models, and stochastic volatility.

Empirical results from both approaches demonstrate that the Gaussian assumption is strongly violated in financial returns data. Residual diagnostics reveal substantial heavy tails and volatility clustering, which are not adequately captured by constant-variance Gaussian models. Incorporating heavy-tailed innovations through the Student- t VAR significantly improves distributional fit and predictive performance.

Among all models considered, the mixture VAR achieves the best overall performance. By explicitly modeling regime switching, it captures both normal and high-volatility market conditions, leading to improved predictive accuracy and more realistic tail risk estimates. The stochastic volatility VAR successfully captures time-varying variance but is less competitive in predictive scoring compared to models that directly account for tail behavior.

Importantly, these conclusions are consistent across both point estimation and Bayesian inference. While the two approaches differ in how parameters are estimated, they produce similar qualitative rankings of model performance. This suggests that the benefits of non-Gaussian innovations are robust to the choice of estimation method.

Key Takeaways

The results highlight several key insights. First, accounting for heavy tails is essential for accurate probabilistic modeling of financial returns. Even relatively simple extensions such as the Student- t VAR provide substantial improvements over Gaussian models.

Second, modeling structural changes in volatility through regime switching yields the strongest gains in predictive performance. The mixture VAR effectively captures rare but extreme events, which are critical for both forecasting and risk management.

Third, different modeling approaches address distinct aspects of the data: heavy-tailed models improve robustness to outliers, mixture models capture discrete regime shifts, and stochastic volatility models account for continuous changes in variance. The choice of model therefore depends on the primary objective, whether it is prediction, risk estimation, or interpretability.

Finally, comparing point estimation and Bayesian inference highlights an important methodological trade-off. Point estimation methods are computationally efficient and straightforward to implement, making them suitable for large-scale or real-time applications. In contrast, Bayesian methods incorporate parameter uncertainty and produce smoother predictive distributions, leading to more stable risk estimates in some cases. Despite these differences, both approaches consistently show that flexible, non-Gaussian models outperform the Gaussian benchmark.

Overall, the findings emphasize the importance of flexible probabilistic modeling in financial time series and demonstrate that relaxing Gaussian assumptions leads to substantial gains in both predictive accuracy and risk assessment.

9 Limitations

Modeling Assumptions

Despite the flexibility introduced by non-Gaussian innovations, the models considered still rely on several simplifying assumptions. All specifications are based on a linear VAR framework, which may fail to capture nonlinear dependencies or more complex interactions between assets. Additionally, the Student- t model assumes a fixed degrees-of-freedom parameter, which restricts the adaptability of tail behavior over time.

The mixture VAR imposes a fixed number of regimes, requiring the number of components to be specified a priori. This may lead to model misspecification if the true data-generating process involves a more complex or continuous structure. Similarly, the stochastic volatility model assumes independent latent volatility processes across assets, ignoring potential co-movement in volatility dynamics.

Computational Constraints

The more flexible models introduce significant computational overhead. Bayesian estimation via MCMC is computationally intensive, particularly for the mixture and stochastic volatility models, which involve latent variables and high-dimensional parameter spaces. This limits the number of iterations and simulation draws that can be practically used, potentially affecting convergence and precision. Additionally, rolling forecast evaluation requires repeated model estimation, further increasing computational cost. As a result, certain design choices, such as reduced simulation size or evaluation frequency, were necessary to keep the analysis tractable.

Data Limitations

The analysis is based on a relatively small set of four equity ETFs, which, while representative of broad market movements, may not fully capture the diversity of financial assets. The focus on daily returns also limits the ability to study intraday dynamics or higher-frequency effects. Moreover, the dataset is restricted to historical observations, and model performance may differ under future market conditions or structural changes not present in the sample. External factors such as macroeconomic variables, interest rates, or geopolitical events are not explicitly incorporated, which may limit the explanatory power of the models.

10 Discussion

Implications for Financial Modeling

The empirical results highlight the importance of moving beyond Gaussian assumptions in financial time series modeling. Standard VAR frameworks, while widely used, fail to capture key features of asset returns, particularly heavy tails, volatility clustering, and regime shifts. Incorporating non-Gaussian innovations leads to substantial improvements in both predictive performance and risk estimation.

From a practical standpoint, the results suggest that models used for forecasting and risk management should explicitly account for extreme events. The superior performance of the mixture VAR indicates that regime-switching behavior is a critical component of financial markets, particularly during periods of stress. At the same time, the strong performance of the Student- t model demonstrates that even relatively simple heavy-tailed extensions can provide meaningful gains.

These findings reinforce the view that financial returns are best understood through flexible probabilistic models that allow for deviations from classical assumptions. In particular, the ability to model tail risk and time-varying uncertainty is essential for applications such as portfolio management, stress testing, and financial regulation.

Extensions and Future Work

Several directions for future research emerge from this work. First, the mixture VAR framework could be extended to allow for a data-driven number of regimes, for example through Bayesian nonparametric methods such as Dirichlet process mixtures. This would remove the need to pre-specify the number of components and allow for more flexible modeling of market structure.

Second, the stochastic volatility model could be extended to incorporate multivariate volatility dynamics, allowing for co-movement in volatility across assets. More advanced specifications, such as factor stochastic volatility or Wishart processes, may provide a richer representation of joint risk.

Third, hybrid models that combine multiple sources of flexibility—such as heavy-tailed innovations with stochastic volatility or regime switching—may further improve performance. These models could capture both discrete and continuous changes in market conditions.

Finally, incorporating additional information, such as macroeconomic variables or high-frequency data, may enhance predictive performance and provide a more comprehensive view of financial dynamics. From a probabilistic machine learning perspective, integrating these models with modern approaches such as variational inference or deep generative models represents a promising direction for scalable and flexible time series modeling.

11 Contribution Statement

- **Sijia Zhu** was primarily responsible for the theoretical development and literature review. This included synthesizing prior work on Bayesian VAR models, non-Gaussian innovations, and forecast evaluation methods, as well as developing the theoretical framework that underpins the models analyzed in this study.
- **Sibo Tao** contributed to the overall project direction and provided the dataset used in the analysis. In addition, he implemented the point estimation framework, including Gaussian, Student- t , and mixture-based approaches, and contributed to the corresponding empirical results.
- **Jonathan Ma** was responsible for data preprocessing, including cleaning, transformation, and construction of return series. He implemented the core modeling pipeline, including Bayesian estimation via MCMC, and developed the analytical pipeline for model comparison. This included coding the forecasting procedures, evaluation metrics, and plotting infrastructure. He also managed version control and project codebase through GitHub, and resolving merge conflicts when needed. He also built the slide deck mirrored off the final report.

All authors contributed to the interpretation of results and the preparation of the final manuscript.

Appendix: Figures

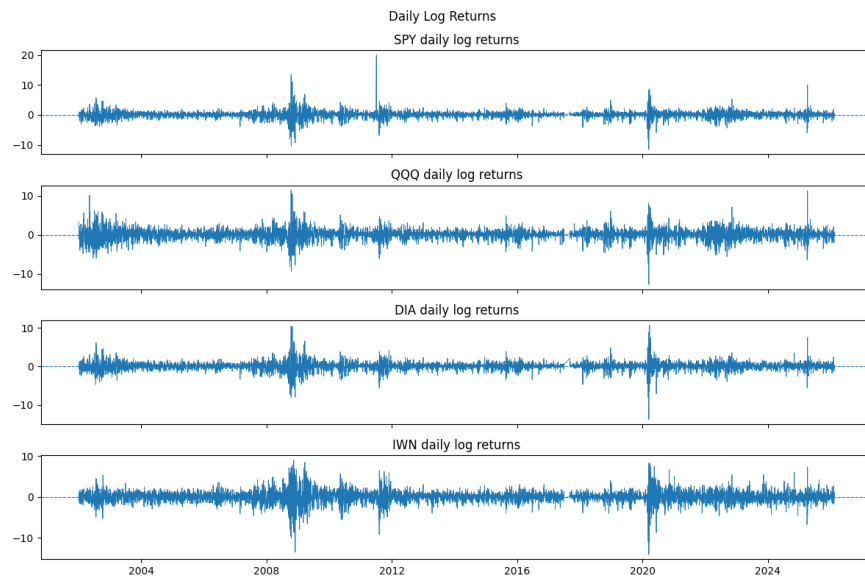


Figure A1: Daily log returns for SPY, QQQ, DIA, and IWN from 2002 to 2026.

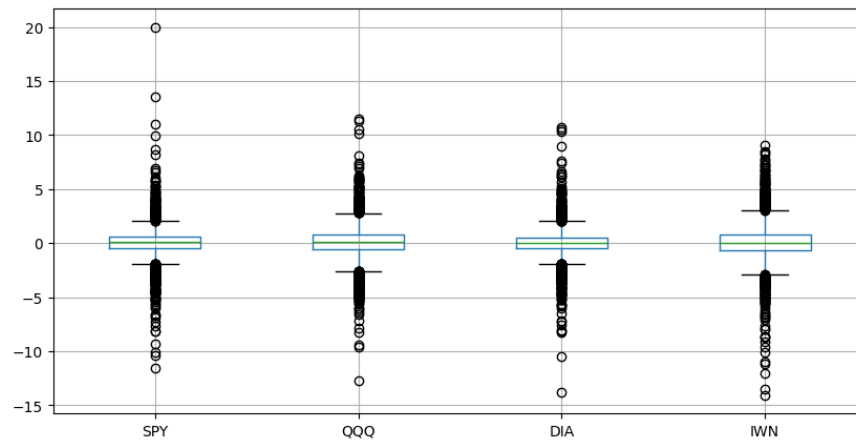


Figure A2: Boxplot of returns showing heavy-tailed behavior

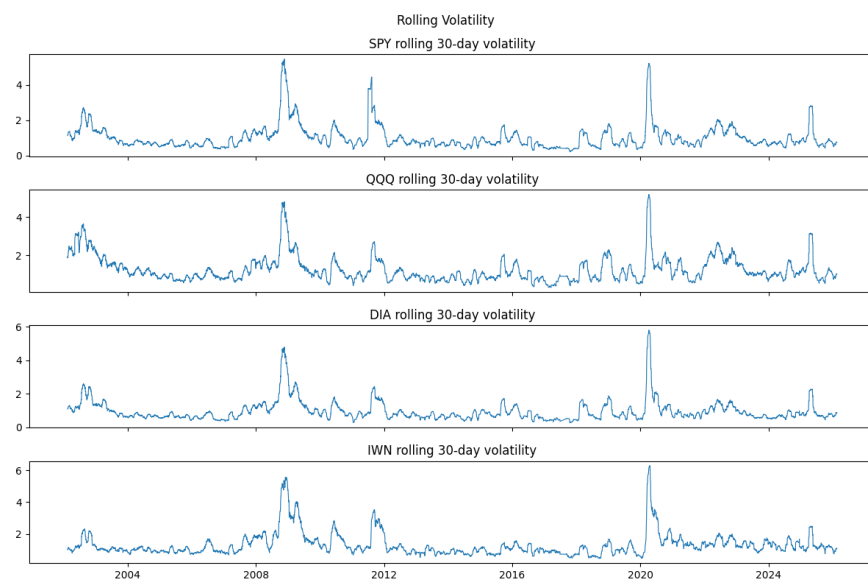


Figure A3: Rolling volatility of returns

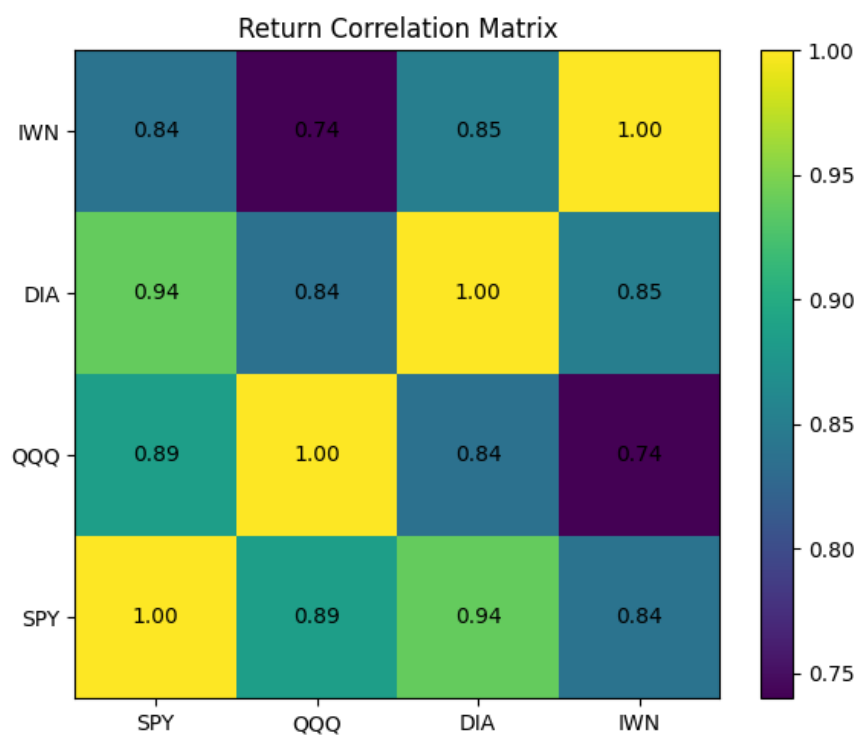


Figure A4: Correlation heatmap of asset returns

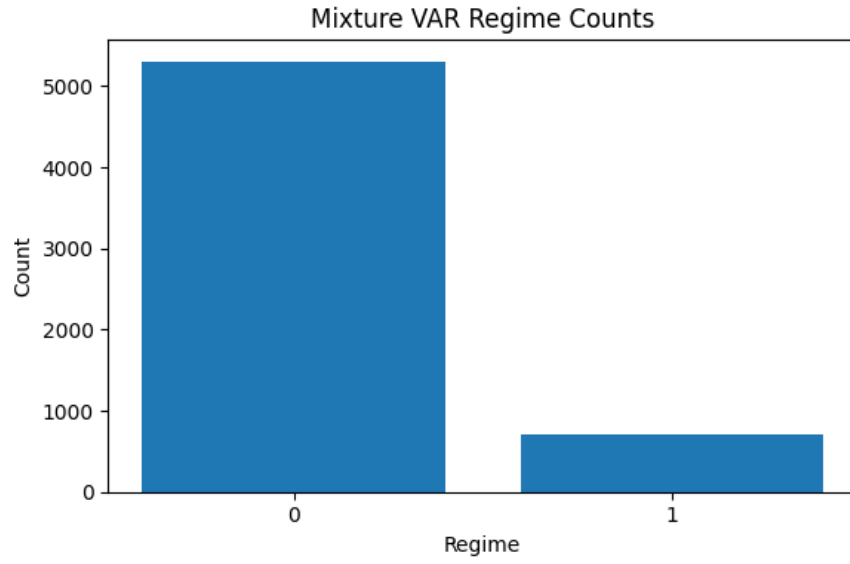


Figure A5: Mixture Regimes for MVAR

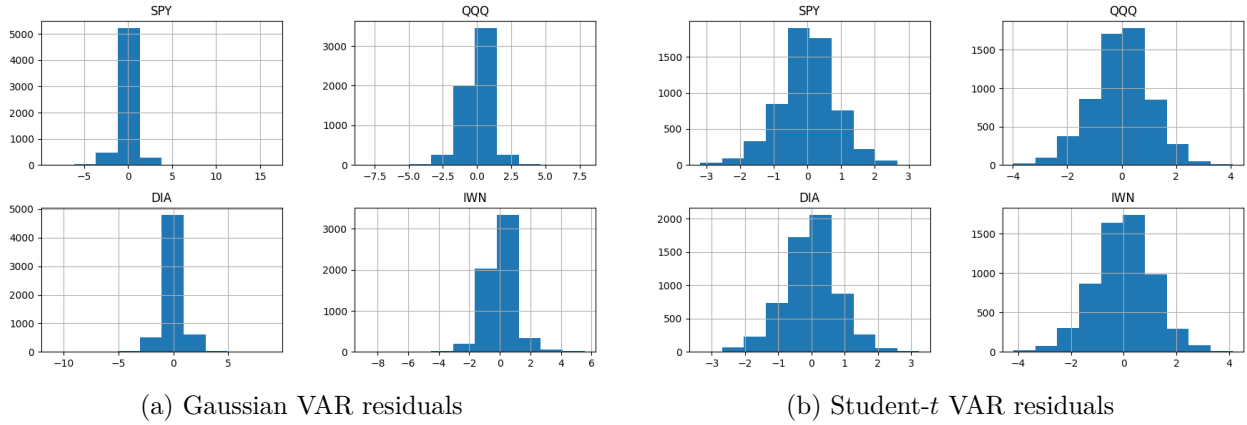


Figure A6: Comparison of residual distributions. The Gaussian VAR exhibits heavier tails, while the Student- t model produces more concentrated residuals after accounting for extreme observations.

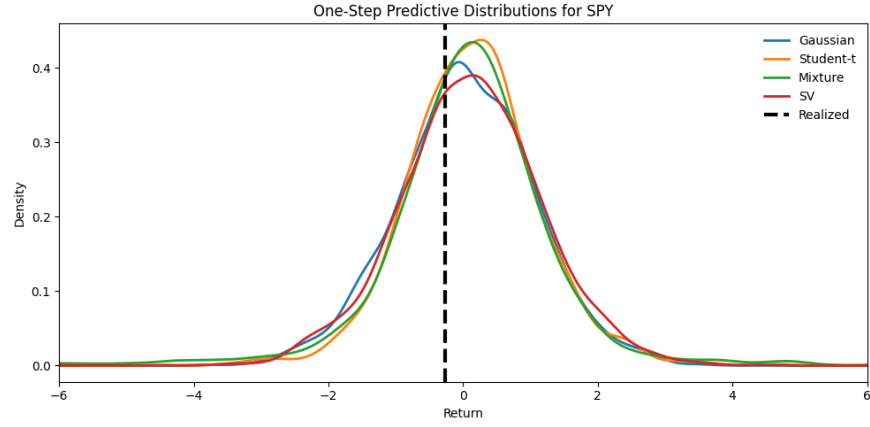


Figure A7: One-step-ahead predictive distributions across models.

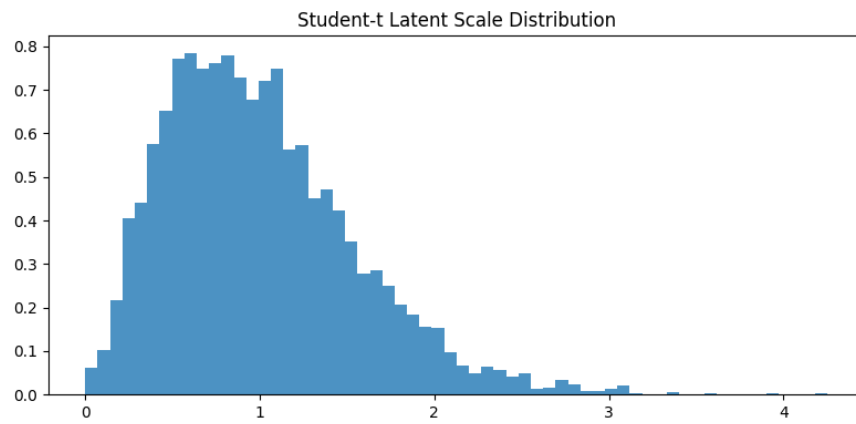


Figure A8: Latent scale parameter λ_t in the Student- t VAR. Lower values correspond to down-weighting of extreme observations, allowing the model to adapt to heavy-tailed behavior.

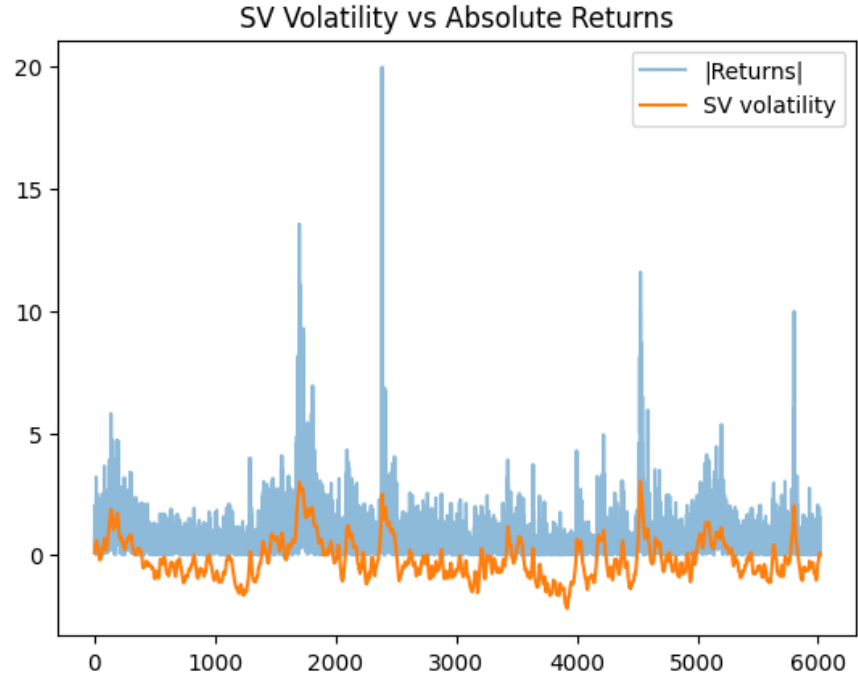


Figure A9: Posterior mean latent volatility paths from the SV-VAR model.

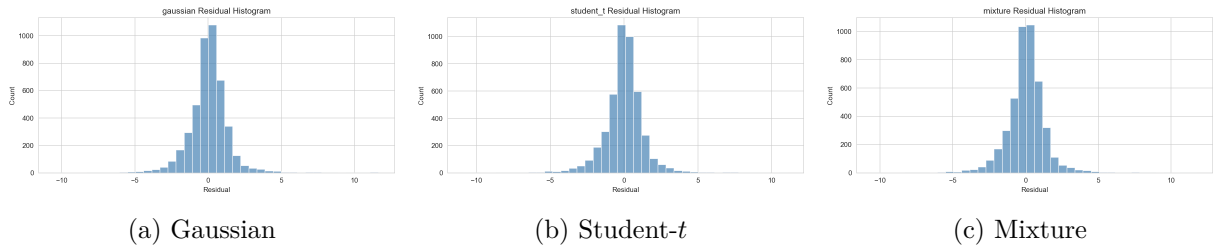


Figure A10: Residual distributions under point estimation.

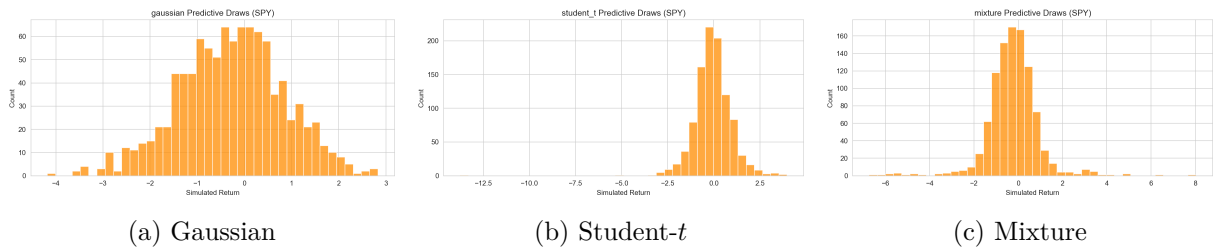
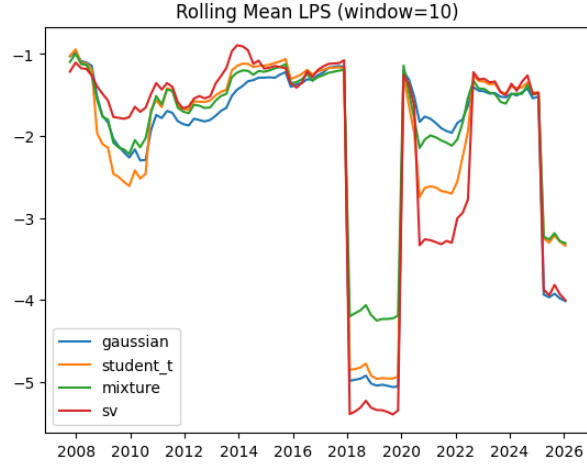
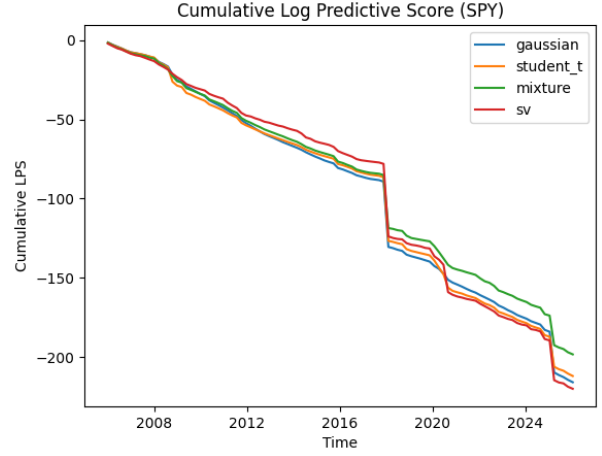


Figure A11: Predictive distributions under point estimation.

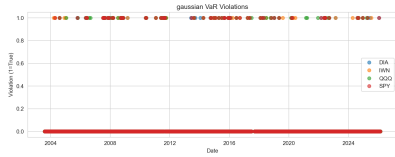


(a) Rolling LPS

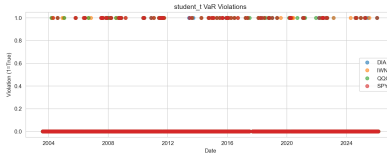


(b) Cumulative LPS

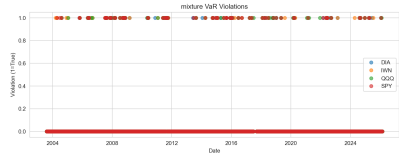
Figure A12: Time-varying predictive performance across models. Rolling LPS highlights short-term forecast accuracy, while cumulative LPS summarizes overall performance, with the mixture model consistently outperforming alternatives.



(a) Gaussian



(b) Student- t



(c) Mixture

Figure A13: VaR violations under point estimation.

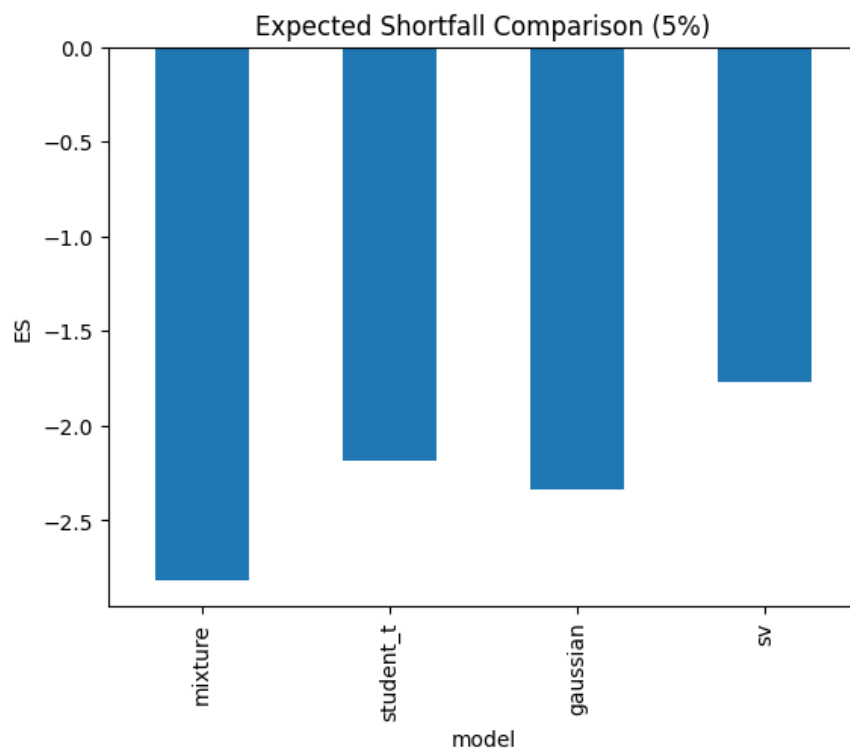


Figure A14: Comparison of Expected Shortfall (ES) across models at the 5% level.

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